

YFM FUSION COMPANION

Illustrated User Guide

Yu-Gi-Oh! Forbidden Memories - Windows x64 companion

Manual fusion tools work without an emulator

Optional read-only live and save integration for NTSC-U RetroArch/SwanStation

Independent fan-made utility - offline artwork with separate attribution

Install and start

Use the release ZIP - GitHub's source archive is for developers

Four steps

- Download a Windows x64 ZIP from the latest release on the Releases page.
- Extract the entire ZIP. Never run the program from inside the ZIP preview.
- Keep Resources, Documentation and Licenses beside the executable. Resources contains the database, artwork and research.
- Start YFM Fusion Companion.exe. The bottom line should report 722 cards and 25,146 resolved fusion pairs.

What is included

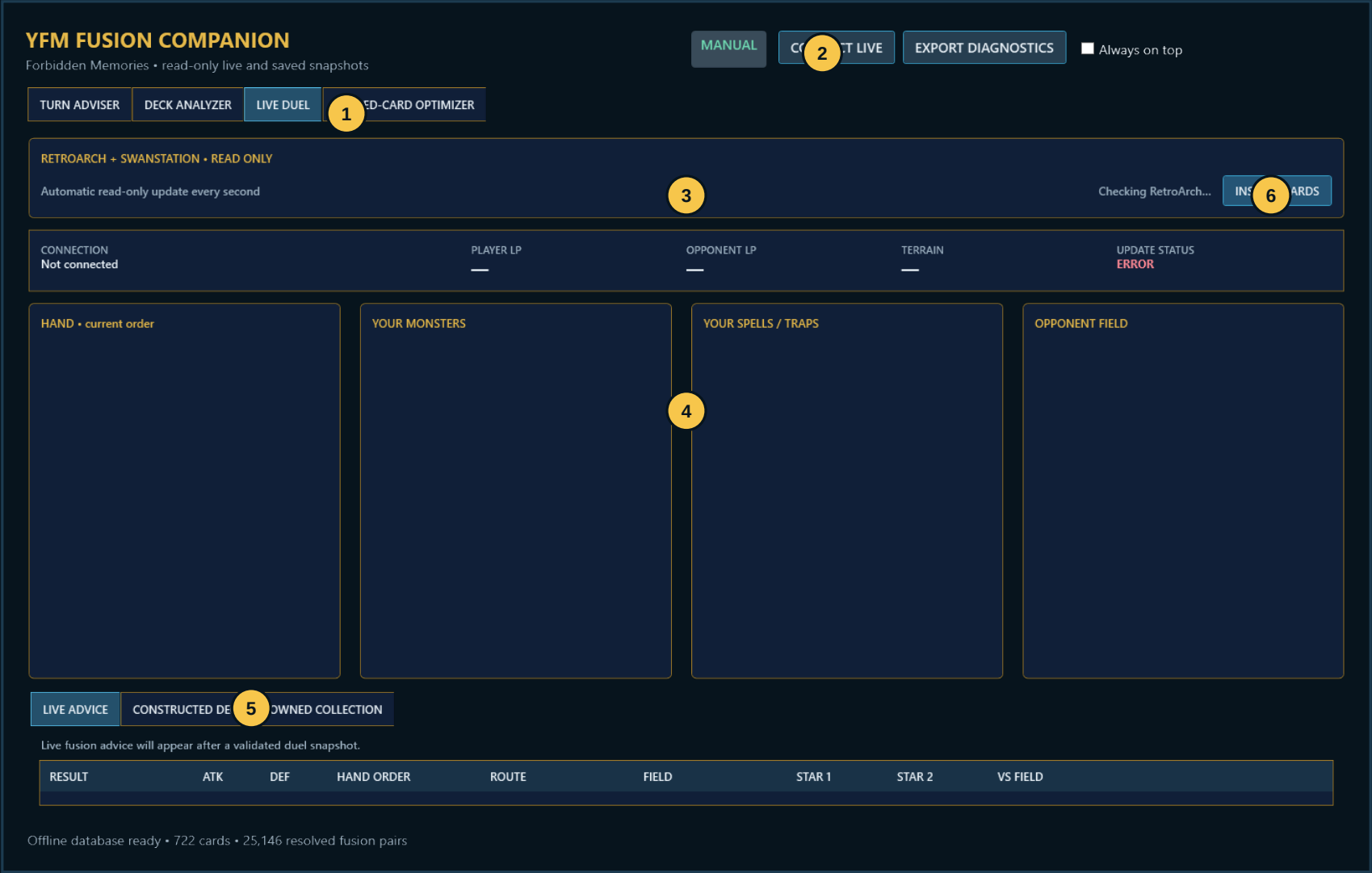
- The portable program is self-contained. Ordinary users do not install .NET.
- RetroArch/SwanStation is optional. It is only needed for Live Duel and automatic save discovery.
- Windows 10 or 11 x64 is required. The release is not code-signed, so SmartScreen may appear.
- Documentation includes the illustrated manual, searchable design guide and a source-code index. See CONTRIBUTING.md online to build from source.

Optional RetroArch setup

- RetroArch: Settings > Network > Network Commands = ON; Network Command Port = 55355.
- Leave Network RetroPad and stdin Commands OFF. Restart RetroArch once, then launch the NTSC-U game with SwanStation.
- The companion connects to 127.0.0.1 and sends read-only status and memory-read requests. Keep unsolicited inbound UDP 55355 blocked in Windows Firewall.

Interface map

The full window keeps source, display, troubleshooting, and workspace controls visible



- 1

Workspace tabs select the job.
- 2

Source badge and Compact Live are global.
- 3

Read-only connection and summary.
- 4

Hand and player/opponent field tables.
- 5

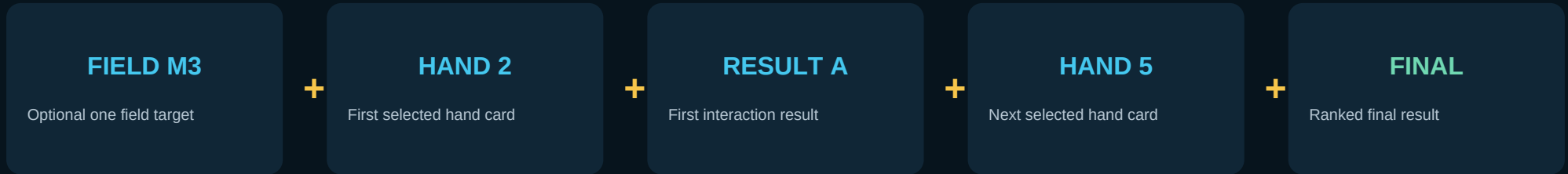
Advice, deck, and collection subtabs.
- 6

Inspector opens only when requested.

Turn Adviser

Enter only cards that actually exist in the current hand and player field

Legal selection order



Inputs and controls

- Hand: up to five ordered cards.
- Monster Field and Spell/Trap Field: up to five optional positions each.
- Analyze Turn ranks legal routes by final ATK, then DEF.
- Tab accepts the top autocomplete result and moves to the next input.
- Empty slots are ignored; duplicate copies remain distinct by slot.

Rules the route obeys

- The field card, when used, interacts with the first selected hand card.
- Only one occupied field position may participate.
- Intermediate results continue through later hand cards in the displayed order.
- Non-fusions that merely discard a material are omitted.
- Compatible equips are terminal; their bonus cannot survive a later monster fusion.
- Documented glitches can be included or excluded explicitly.

Deck Analyzer

Every physical five-card hand is examined for a complete 40-card deck

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

OWNED-CARD OPTIMIZER

DECK • up to 40 cards

0 / 40 cards • add cards below or load your saved deck

Add a card • name or number

Empty slot • optional

ADD COPY

REMOVE SELECTED COPY

ADVANCED • EDIT INDIVIDUAL DECK SLOTS

ANALYZE

LOAD CURRENT L

CANCEL

CLEAR

Include documented glitch fusions

The analyzer reads the current validated save when you load the deck.

ANY FUSION / EQUIP

ATK ≥ 2500

ATK ≥ 2800

ATK ≥ 3000

EXPECTED BEST ATK

DEAD HAND

Fill any number of slots. A 40-card deck examines exactly 658,008 physical five-card hands.

RESULT

FINAL ATK

FINAL DEF

HAND CHANCE

HANDS

REPRESENTATIVE ROUTE

GLITCH

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Enter up to 40 cards.

2

Load the validated saved deck in one click.

3

Run, cancel, clear, and glitch controls.

4

Exact probability and expected-ATK metrics.

5

Per-result hand chance and representative route.

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Live Duel

Validated state refreshes automatically every second

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

OWNED-CARD OPTIMIZER

RETROARCH + SWANSTATION • READ ONLY

Automatic read-only update every second

Checking RetroArch... INSARDS

CONNECTION
Not connected

PLAYER LP
—

OPPONENT LP
—

TERRAIN
—

UPDATE STATUS
ERROR

HAND • current order

YOUR MONSTERS

YOUR SPELLS / TRAPS

OPPONENT FIELD

LIVE ADVICE

CONSTRUCTED DECK

OWNED COLLECTION

Live fusion advice will appear after a validated duel snapshot.

RESULT

ATK

DEF

HAND ORDER

ROUTE

STAR 1

STAR 2

VS FIELD

Offline database ready • 722 cards • 25,146 resolved fusion pairs

- 1

Connection and read-only status.
- 2

Life Points, terrain, and update health.
- 3

Current ordered five-card hand.
- 4

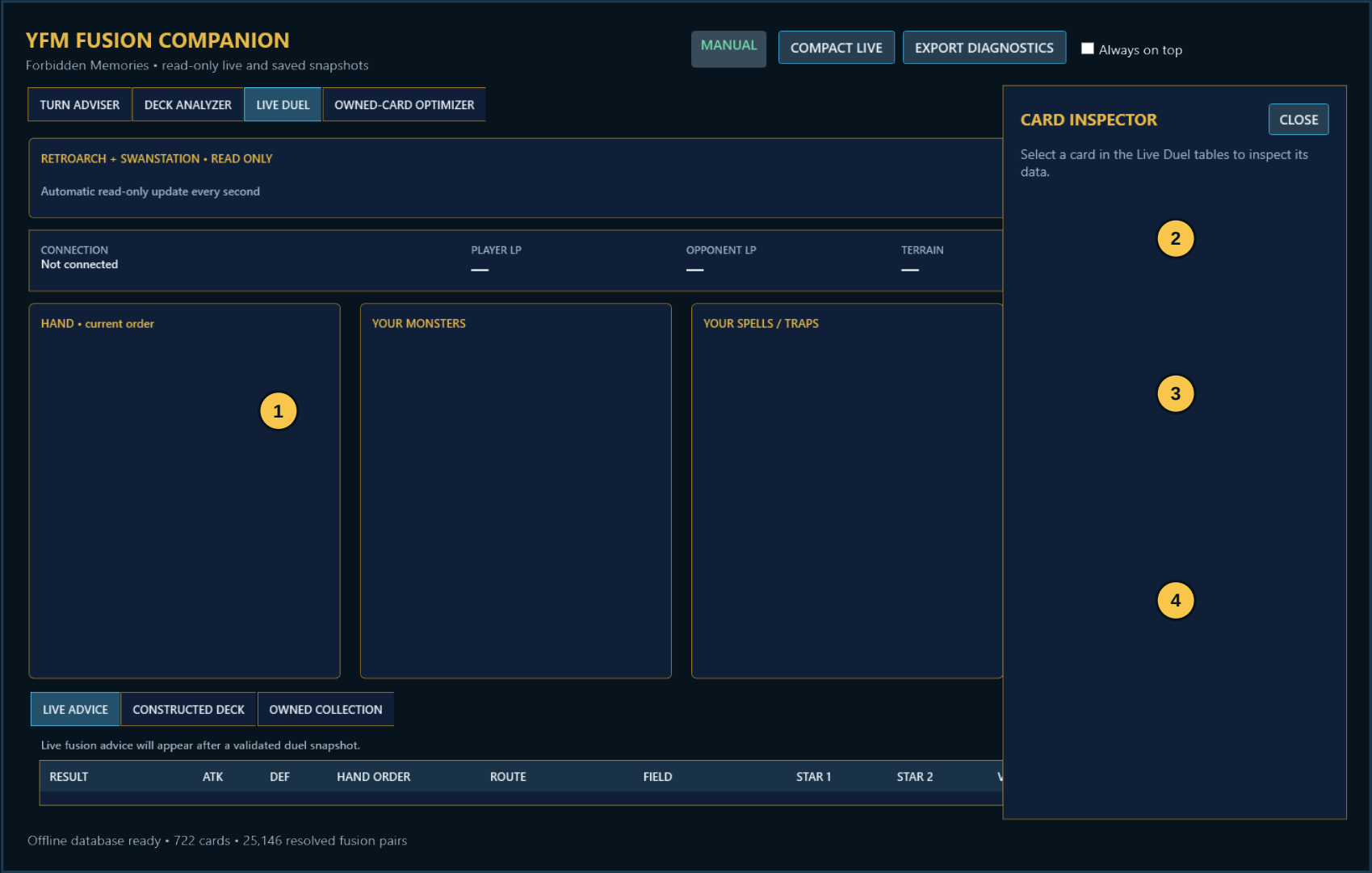
Player and opponent active field positions.
- 5

Best legal live routes and both guardian-star chains; F#? preserves unknown opponent state.
- 6

Open the selected-card inspector.

Card Inspector

Select a live card, then inspect basic and advanced catalog data



- 1

Select a card in any live table.
- 2

Inspector slides over the right edge.
- 3

Basic type, ATK, and DEF appear first.
- 4

Advanced Data expands catalog relationships.

Recommended campaign decks

Six researched reference builds; opening-hand availability is not a duel win rate

DECKS FOR THE JOURNEY

Six community-informed strategies for the original game. Choose an icon to explore the full 40-card build.

MANUAL COLLECTION • isolated rendering fixture • Counts match required copies, including cards in your saved deck and chest.



1 Thunder Tide
Campaign → endgame with power-ups

40 / 40 required copies owned



Meteor Crossfire
Late campaign / final gauntlet

23 / 40 copies owned



Mercury Eclipse
Endgame alternative to Thunder

12 / 40 copies owned



Iron & Meteor
Farmed endgame / fewer materials

11 / 40 copies owned



Sand & Sorcery
Fan synergy → equip-heavy endgame

9 / 40 copies owned



Crimson Toolbox
Fan fusion web → upgraded endgame

14 / 40 copies owned

Thunder Tide • Campaign → endgame with power-ups

Repeatable Thunder fusions, a single Umi, compatible power-ups and reliable removal.

40 / 40 required copies owned • 0 missing copies

HOW TO PLAY AND UPGRADE

Start with available Dragon/Thunder materials. Replace weak three-material chains with **3** per two-card recipes as you collect them. Umi + Megamorph + Dragon Treasure takes Twin-headed Thunder Dragon to 4,800 ATK over separate plays. Preserve Raigeki for threats you can't currently beat.

Crawling Dragon + Electric Snake → Twin-headed Thunder Dragon
Baby Dragon + Thunder Dragon → Twin-headed Thunder Dragon

A 2,800 ATK fusion alone is insufficient late in the game. The displayed setup odds require the field and equips to be present in the same opening hand; they do not assume an active Umi or predict a win.

CHECK ALL **4** / 40 HANDS

CANCEL CHECK

Check the reference deck to see its opening-hand setup availability.

These metrics describe the complete reference list, including cards you may **6** sing. Setups may require several turns. Opponent fields, guardian stars, equips and responses can change the outcome.

CARD	NEED	OWN	MISSING	ROLE
Crawling Dragon	3	3	0	Fusion materials

ADAPT THIS STRATEGY **5** / COLLECTION

CLOSE

Uses owned cards as a starting deck; missing cards are replaced. Other strategies are still compared, so the best result can differ.

- 1 Distinct icons select each reference build.
- 2 Required copies already owned appear out of 40 beneath each icon.
- 3 Missing cards and progression advice explain the strategy.
- 4 Check all opening hands; cancel whenever needed.
- 5 Adapt a strategy to your collection; the optimizer compares other starts too.
- 6 Reference metrics can include missing cards; read sources and limits.

Suggested deck and ordering

Save import is integrated into Deck Analyzer and Owned-Card Optimizer

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

Starting strategy: automatic • Compare all six builds: [YFM FUSION COMPANION](#)

exact analysis of 658,008 hands • 40/40 cards • seed 5850701 • ≥3500 setup available 0.0% • board clear drawn 0.0% • >4500 setup 0.0% • strong body 33.8% • no monster 0.0% • highest setup 3,000 ATK. Setups may require separate turns; not win probabilities. • support for Blue-eyes White Dragon: 0 useful field / 0 compatible equips • below the 1-field / 2-equip structure target.

OWNED COLLECTION

All types

Name

Add owned card

Empty slot • optional

ADD CARD

CUSTOM ART FOLDER...

CUSTOM SELECTED ART...

13 shown • 13 distinct • 39 copies

Baby Dragon

Dragon • ATK 1,200 / DEF 700 • ♀ / ♂

Owned 3 • manual quantity

Proposed deck: 3x

Black-lance Dragon

Dragon • ATK 1,500 / DEF 800 • ♀ / ♂

Owned 3 • manual quantity

Proposed deck: 3x

Blue-eyes White Dragon

Dragon • ATK 3,000 / DEF 2,500 • ♂ / ♀

Owned 3 • manual quantity

Proposed deck: 3x

Feral Imp

Fiend • ATK 1,300 / DEF 1,400 • ♂ / ♀

Owned 3 • manual quantity

Proposed deck: 3x

ATK ≥ 2800

0.00 %

ATK ≥ 2500

0.00 %

EXPECTED BEST

1,582

DEAD HAND

14.93 %

General campaign safety (33 opponents; final gauntlet excluded) • best found unless proof status explicitly says otherwise.

Modeled answers for 28/33 opponents • weakest modeled matchup 620 • opening answer coverage 83.94%. These are modeled safety scores, not duel-win probabilities.

Virtual purchase plan uses 9,005,806 of 0 Star Chips. Password redemption history is not stored by the game.

DECK

OUTCOMES

LIMITS

LOW VALUE

STAR CHIPS

THREATS

CARD ORDER

2

Alphabetical

3

COPIES	CARD	NO.	OWNED	BUY	TYPE	ATK	DEF	WHY INCLUDED
3	Baby Dragon	4	0	3	Dragon	1200	700	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 5 compatible deck partners; reaches B. Dragon Jungle King (2,100 ATK). Purchase 3 for 2,999,997 Star Chips.
3	Blackland Fire Dragon	10	0	3	Dragon	1500	800	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 4 compatible deck partners; reaches B. Dragon Jungle King (2,100 ATK). Purchase 3 for 240 Star Chips.
3	Blue-eyes White Dragon	1	0	3	Dragon	3000	2500	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 0 compatible deck partners; base ATK 3,000. Purchase 3 for 2,999,997 Star Chips.
3	Feral Imp	6	0	3	Fiend	1300	1400	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 2 compatible deck partners; reaches Koumori Dragon (1,500 ATK). Purchase 3 for 420 Star Chips.
3	Hitotsu-me Giant	3	0	3	Beast-Warrior	1200	1000	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 2 compatible deck partners; reaches Koumori Dragon (1,500 ATK). Purchase 3 for 310 Star Chips.

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Enter or load owned quantities; card artwork works offline.

2

Alphabetical is the default card order.

3

Choose card number, ATK or DEF; the choice is remembered.

4

ATK and DEF sort highest first; card numbers lowest first.

5

Campaign summary keeps uncertainty visible.

6

Fields, equips and removal are scored as coherent support; setups may need separate turns.

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Compact Live

A narrow right-side companion that leaves the central duel area visible



- 1

Result name and effective ATK.
- 2

Hand routes use plain slot numbers.
- 3

F(1)-F(5) are monster positions; F(6)-F(10) are spell/trap positions.
- 4

20px stars sit beneath Result; warm/red glows clarify choices and known enemy relations.
- 5

PIN stays above the game; FULL restores the normal window.

Compatibility, privacy, and troubleshooting

Compatibility boundary

- Manual adviser, analyzer, and optimizer work from the bundled offline catalog.
- Live decoding and automatic discovery are validated for NTSC-U / SLUS-01411 in RetroArch/SwanStation.
- Other regions, revisions, emulators, cores, and modified games are not claimed compatible; use manual tools.
- RetroAchievements Hardcore Mode works because the companion never uses save states.

Read-only and private by design

- No game input, memory writes, save edits, config edits, telemetry, updater, ads, or online API.
- Local settings store window bounds, compact/topmost choices, and the last selected save path.
- Diagnostics are created only on request and exclude gameplay/card values.
- No ROM, BIOS, emulator or memory-card save is distributed. Bundled artwork has separate attribution.

When something is unavailable

- ERROR or DISCONNECTED: confirm Network Commands, port 55355, firewall scope, SwanStation, and that RetroArch was restarted.
- NOT IN DUEL: connectivity is healthy; enter a duel. Stale duel structures are intentionally hidden in story mode.
- Save looks old: save in-game, close content so the emulator flushes the file, then Refresh in the analyzer or optimizer.
- Wrong region/core: Live Duel may be rejected safely, but every manual feature remains available.
- For support, use Export Diagnostics; review the text yourself before sharing it.